

Midterm Exercise 2: Standing Ovation Problem Model

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Implementing the SOP model was largely a modification of the Schelling model, as even the paper notes. Including multiple neighborhood types and update timings were the most challenging both from a technical perspective and, in the latter case, due to some lack of specificity. For neighborhood types, unlike the Schelling model which was modified to produce this model, I did not see an existing function to get the neighbors of the type I wanted. Instead, I had to either retrieve a subset of the Moore neighborhood from the `get_neighbors()` function or add neighbors to complete the cone neighborhood. Regarding update timing, while the synchronous and asynchronous random updates were relatively trivial to implement, the asynchronous-incentive-based timing was a challenge. First, the description of the model does not specify whether an ordered list should be generated at the start of each “step” in the model or whether the maximally incentivized agent should be re-calculated after each agent updates its status. I opted to first order all agents by incentive and then complete their updates in this order regardless of new incentive values within the step. This therefore requires the addition of a method to calculate their incentive to move so agents can be activated in the proper order. In addition, the paper does not specify how quality signals should be distributed. In some sections of the paper it notes that this would realistically follow some pattern, but due to the lack of specificity of an exact distribution I chose to simply use a uniform random distribution of quality signals between 0 and 1.

Unlike the original model, I did not seem to have stick-in-the-mud behavior as seems to be described in the original paper. In the paper, I understood this behavior to mean that the model had ceased to change, and there were patches or individual agents who had a state contrary to the majority. In my model, however, this seemed to manifest as columns of alternating state. While it is true that the total number of standing individuals had not changed, this did not match my initial interpretation of Smith and Page’s description. This may be more of a communication mismatch than an actual difference in model performance, though. It could also be the case that I did not test a sufficient number of random seed values.

Code GitHub Link:

https://github.com/gabereichman/ABM_Assignments/tree/main/SOP

Resource Usage Statement:

While I did not use any resources for coding help, as I was able to modify the Schelling model to fit the SOP model, I did utilize Gemini to ensure I properly understood the model rules. I uploaded the PDF of the article detailing the model and utilized Gemini by asking questions about the model to clarify my understanding.